

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2019 - 2020)

NEET(UG)
SEMI MAJOR
08-03-2020

PRE-MEDICAL : SPARK COURSE (PHASE - I)*This Booklet contains 24 pages.***Do not open this Test Booklet until you are asked to do so.****Read carefully the Instructions on the Back Cover of this Test Booklet.****Important Instructions :**

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

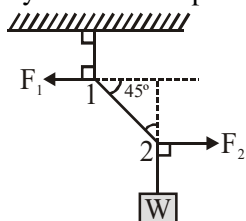
Centre of Examination (in Capitals) :_____

Candidate's Signature :_____ Invigilator's Signature :_____

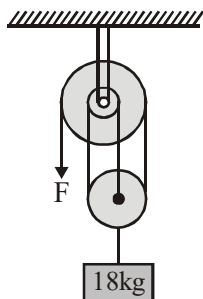
Your Target is to secure Good Rank in Pre-Medical 2020

TOPIC : NLM and Friction, Work, Power and Energy, Collision and centre of mass and Rotational motion, Kinematics, Electrostatics, Thermal Physics, Current Electricity and Heating effect of current, Magnetic effect of current, EMI, AC, EMW

1. In the figure the tension in the string between 1 and 2 is 60 N. Find the magnitude of the horizontal force $\vec{F}_1$ and $\vec{F}_2$ that must be applied to hold the system in the position shown.

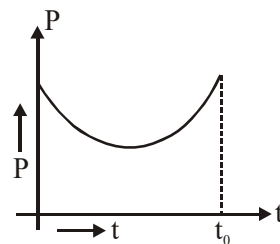


- (1) $|\vec{F}_1| = |\vec{F}_2| = 40\sqrt{2}\text{ N}$
 (2) $|\vec{F}_1| = |\vec{F}_2| = 30\sqrt{2}\text{ N}$
 (3) $|\vec{F}_1| = |\vec{F}_2| = 10\sqrt{2}\text{ N}$
 (4) $|\vec{F}_1| = |\vec{F}_2| = 20\sqrt{2}\text{ N}$
2. In the figure, at the free end of the light string, a force F is applied to keep the suspended mass of 18 kg at rest. Then the force exerted by the ceiling on the system (assume that the string segments are vertical and the pulleys are light and smooth) is: ($g = 10 \text{ m/s}^2$) -

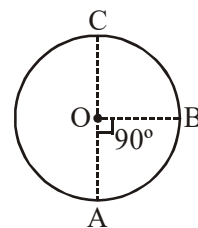


- (1) 60 N (2) 120 N (3) 180 N (4) 240 N
3. A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 50N, When the lift is stationary. If the lift moves downward with an acceleration of 5 m/s^2 , the reading of the spring balance will be :
- (1) 25 N (2) 75 N
 (3) 15 N (4) 49 N

4. Power versus time graph for a given force is given below. Work done by the force upto time $t(\leq t_0)$.



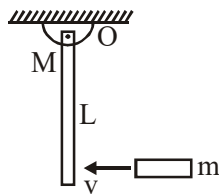
- (1) First decreases then increases
 (2) First increases then decreases
 (3) Always increases
 (4) Always decreases
5. A particle crosses the topmost point C of a vertical circle with critical speed; then the ratio of velocities at points A, B and C is -



- (1) 3 : 2 : 1 (2) 5 : 3 : 1
 (3) $5^2 : 3^2 : 1^2$ (4) $\sqrt{5} : \sqrt{3} : \sqrt{1}$
6. A particle is released from a height H . At certain height its kinetic energy is two times its potential energy. Height and speed of particle at that instant are -

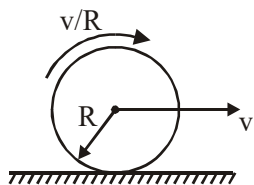
- (1) $\frac{H}{3}, \sqrt{\frac{2gH}{3}}$ (2) $\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$
 (3) $\frac{2H}{3}, \sqrt{\frac{2gH}{3}}$ (4) $\frac{H}{3}, \sqrt{2gH}$

7. A wooden log of mass M & length L is hinged by a frictionless nail at O . A bullet of mass m strikes with velocity v & sticks to it. Find the angular velocity of the system immediately after collision -



- (1) $\frac{mv}{M+m}$ (2) $\frac{3mv}{(M+m)L}$
 (3) $\frac{3mv}{(M+3m)L}$ (4) $\frac{mv}{(M+3m)L}$

8. A disc is performing pure rolling on a smooth stationary surface with constant angular velocity as shown in figure. At any instant, for the lower most point of the disc -

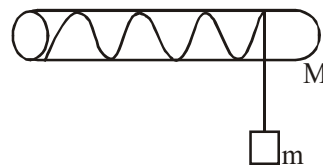


- (1) Velocity is v , acceleration is zero
 (2) Velocity is zero, acceleration is zero
 (3) Velocity is v , acceleration is $\frac{v^2}{R}$
 (4) Velocity is zero, acceleration is nonzero

9. Find the speed V reached by a car of mass ' m ' that is driven with constant power P and X is the distance travelled from rest -

- (1) $\left(\frac{3XP}{m}\right)^{\frac{1}{3}}$ (2) $\left(\frac{2XP}{m}\right)^{\frac{1}{2}}$
 (3) $\left(\frac{XP}{m}\right)^{\frac{1}{3}}$ (4) $\left(\frac{2XP}{3m}\right)^{\frac{1}{4}}$

10. A cord is wound round the circumference of a solid cylinder of radius R and mass M . The axis of the cylinder is horizontal. A weight mg is attached to the end of the cord and falls from rest. After falling through a distance h , the angular velocity of the cylinder will be -



- (1) $\frac{2mg}{M+2m}$ (2) $\sqrt{\frac{2gh}{R}}$
 (3) $\sqrt{\frac{4mgh}{(M+2m)R^2}}$ (4) None of these

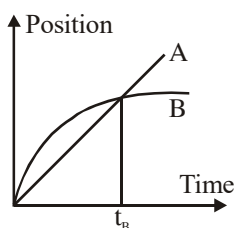
11. A body of mass 2 kg makes an elastic head on collision with another body at rest and continues to move in the same direction with $1/4$ of its initial velocity. Find out mass of the second body.

- (1) 1.2 kg (2) 2 kg
 (3) 4 kg (4) 2.4 kg

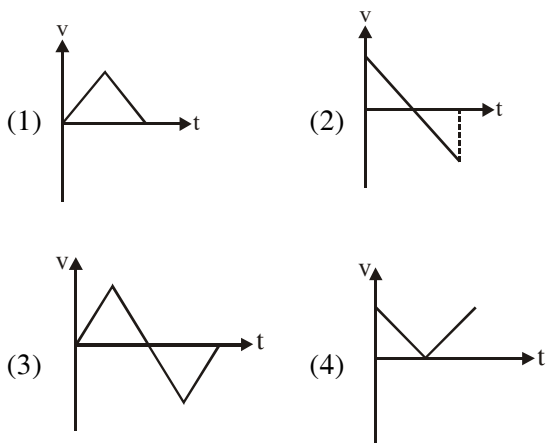
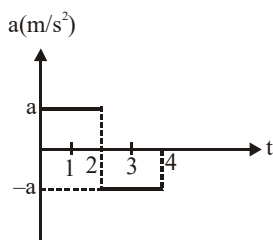
12. If $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ then the direction cosines of the vector $\vec{A}$ are :-

- (1) $\frac{2}{\sqrt{45}}, \frac{4}{\sqrt{45}}$ and $\frac{-5}{\sqrt{45}}$
 (2) $\frac{1}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{3}{\sqrt{45}}$
 (3) $\frac{4}{\sqrt{45}}, 0$ and $\frac{4}{\sqrt{45}}$
 (4) $\frac{3}{\sqrt{45}}, \frac{2}{\sqrt{45}}$ and $\frac{5}{\sqrt{45}}$

13. The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?



- (1) At time t_B , both trains have the same velocity
 (2) Both trains have the same velocity at some time after t_B
 (3) Both trains have the same velocity at some time before t_B .
 (4) Somewhere on the graph, both trains have the same acceleration.
14. A particle starts from rest and undergoes an acceleration as shown in figure. The velocity-time graph from figure will have a shape :-



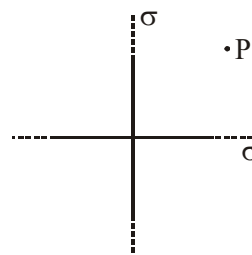
15. Two paper screens A and B are separated by 150 m. A bullet pierces A and B. The hole in B is 15 cm below the hole in A. If the bullet is travelling horizontally at the time of hitting A, then the velocity of the bullet at A is ($g = 10 \text{ ms}^{-2}$) :-

- (1) $100\sqrt{3} \text{ ms}^{-1}$ (2) $200\sqrt{3} \text{ ms}^{-1}$
 (3) $300\sqrt{3} \text{ ms}^{-1}$ (4) $500\sqrt{3} \text{ ms}^{-1}$

16. A car is going in south with a speed of 5 m/s. To a man sitting in car a bus appears to move towards west with a speed of $2\sqrt{6} \text{ m/s}$. What is the actual speed of the bus ?

- (1) 4 ms^{-1} (2) 3 ms^{-1}
 (3) 7 ms^{-1} (4) None of these

17. Two infinitely long uniformly charged non conducting sheets of charge density $\sigma = \sqrt{2} \epsilon_0 \times 10^{-3}$ each are placed perpendicular to each other as shown in figure. What is the resultant electric field intensity at point P?

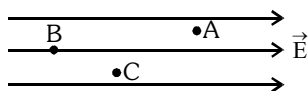


- (1) 10^{-3} N/C (2) $2 \times 10^{-3} \text{ N/C}$
 (3) $\sqrt{2} \times 10^{-3} \text{ N/C}$ (4) 1 N/C

18. Two identical particles each having a charge of $10 \mu\text{C}$ are tied with two strings of equal length 1m at a common point. If both the strings makes an angle of 30° with vertical, then mass of each particle will be :-

- (1) 0.1 kg (2) 1 kg
 (3) 0.15 kg (4) 0.5 kg

19. A, B and C are three points in a uniform electric field. The electric potential is :-



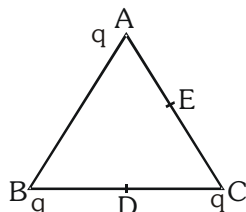
- (1) same at all the three points A, B and C
 (2) maximum at A
 (3) maximum at B
 (4) maximum at C
20. The electric potential V at any point (x, y, z) . (all in metres) in space is given by $V = 4x^2$ volts. The electric field at the point $(1, 0, 2)$ in volt/meter, is :-
- (1) 8 along negative X-axis
 (2) 8 along positive X-axis
 (3) 16 along negative X-axis
 (4) 16 along positive X-axis
21. Three charges each $+q$ are placed at the three corners of an isosceles triangle ABC with sides BC and AC each equal to $2a$. D and E are the mid points of BC and CA respectively. The work done in taking a charge Q from D to E is :-

(1) $\frac{3qQ}{4\pi\epsilon_0 a}$

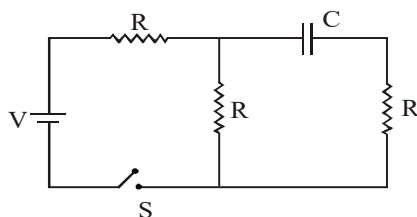
(2) $\frac{3qQ}{8\pi\epsilon_0 a}$

(3) $\frac{qQ}{4\pi\epsilon_0 a}$

(4) Zero



22. Current drawn by the battery just after switch 'S' is closed is :-



- (1) $\frac{2V}{3R}$ (2) $\frac{3V}{2R}$ (3) $\frac{2V}{R}$ (4) $\frac{V}{R}$

23. Work done per mol in an isothermal change is:

(1) $RT \log_{10} \frac{V_2}{V_1}$ (2) $RT \log_{10} \frac{V_1}{V_2}$

(3) $RT \log_e \frac{V_2}{V_1}$ (4) $RT \log_e \frac{V_1}{V_2}$

24. An ideal refrigerator has a freezer at a temperature of -13°C . The coefficient of performance of the engine is 5. The temperature of the air (to which heat is rejected) will be:

- (1) 325°C (2) 325K
 (3) 39°C (4) 320°C

25. A lead bullet of unknown mass is fired with a speed of 180 ms^{-1} into a tree in which it stops. Assuming that in this process two-third of heat produced goes into the bullet and one-third into wood. The temperature of the bullet rises by: (Specific heat of lead = $120 \text{ J/kg}^\circ\text{C}$):

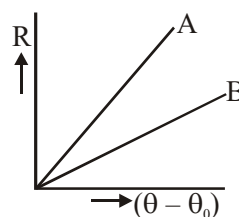
- (1) 140°C (2) 106°C (3) 90°C (4) 100°C

26. One mole of an ideal gas requires 207J heat to raise the temperature by 10K when heated at constant pressure. If the same amount of gas is heated at constant volume to raise the temperature by the same 10K , the heat required is:

- (1) 198.7J (2) 29J
 (3) 215.3J (4) 124J

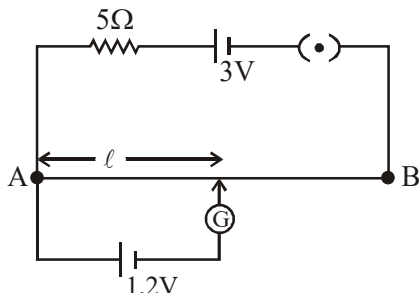
27. Two circular discs A and B with equal radii and equal mass are blackened. They are heated to same temperature and are cooled under identical conditions. What inference do you draw from their cooling curves?

R is rate of cooling:

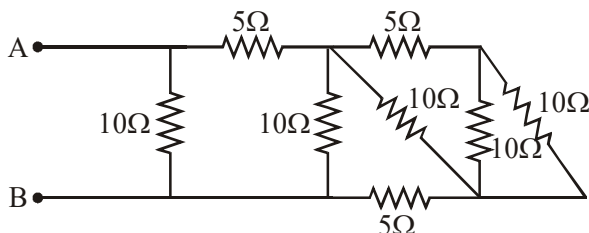


- (1) A and B have same specific heats
 (2) Specific heat of A is less
 (3) Specific heat of B is less
 (4) Nothing can be said

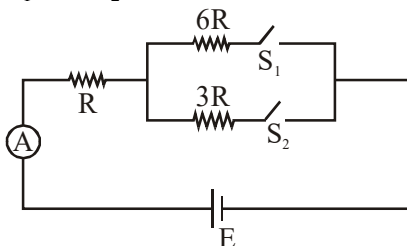
28. The potentiometer wire of length 100 cm has a resistance of $10\ \Omega$. It is connected in series with a resistance of $5\ \Omega$ and an accumulator of emf 3V having negligible resistance. A source of 1.2 V is balanced from point A :



- (1) 60 cm (2) 40 cm (3) 30 cm (4) None
29. Equivalent resistance between terminals A and B

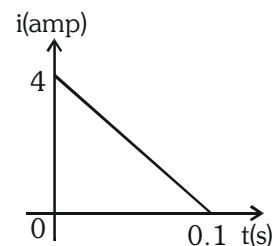


- (1) $2.5\ \Omega$ (2) $15\ \Omega$ (3) $5\ \Omega$ (4) $10\ \Omega$
30. The reading of ammeter is I_1 when only ' S_1 ' is closed and I_2 when only S_2 is closed. The reading is I_3 when both S_1 and S_2 are closed simultaneously then :

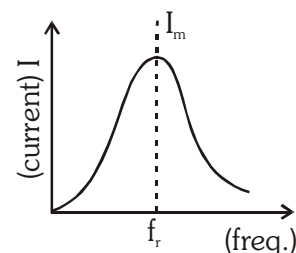


- (1) $I_1 > I_2 > I_3$ (2) $I_3 > I_2 > I_1$
 (3) $I_2 > I_3 > I_1$ (4) $I_3 > I_1 > I_2$
31. A battery of internal resistance $3\ \Omega$ is connected to a variable resistor whose value can vary from $5\ \Omega$ to $10\ \Omega$. The resistance is initially set at $5\ \Omega$, if the resistance is now increased then:
- (1) Power consumed by it will increase
 (2) Power consumed by it will decrease
 (3) Power consumed by it may increase or may decrease
 (4) Power consumed by it first increase then decrease

32. In a coil of resistance $10\ \Omega$, the induced current developed by changing magnetic flux through it, is shown in figure as a function of time. The magnitude of change in flux through the coil in Weber is:-

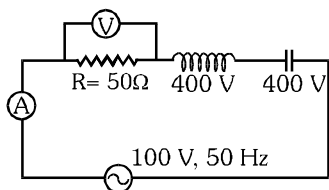


- (1) 6 (2) 4 (3) 8 (4) 2
33. A wire loop is rotated in magnetic field about the diameter. The frequency of change of direction of the induced e.m.f. is :
- (1) Six times per revolution
 (2) Once per revolution
 (3) twice per revolution
 (4) four times per revolution
34. A transformer having efficiency of 90% is working on 200V and 3kW power supply. If the current in the secondary coil is 6A, the voltage across the secondary coil and the current in the primary coil respectively are:-
- (1) 300 V, 15A (2) 450 V, 15A
 (3) 450V, 13.5A (4) 600V, 15A
35. The graph shows variation of I with f for a series R-L-C network. Keeping L and C constant. If R decreases :



- (a) Maximum current (I_m) increases
 (b) Sharpness of the graph increases
 (c) Quality factor increases
 (d) Band width increases
- (1) a, b, c (2) b, c, d (3) c, d, a (4) All

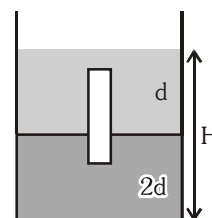
36. If the current through an inductor of inductance L is given by $I = I_0 \sin \omega t$, then the voltage across inductor will be :-
 (1) $I_0 \omega L \sin (\omega t - \pi/2)$ (2) $I_0 \omega L \sin (\omega t + \pi/2)$
 (3) $I_0 \omega L \sin (\omega t - \pi)$ (4) None of these
37. In given LCR circuit, the voltage across the terminals of a resistance & current will be-



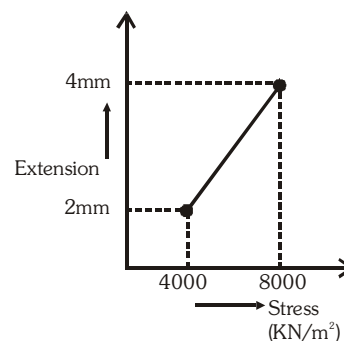
- (1) 400V, 2A (2) 800V, 2A
 (3) 100V, 2A (4) 100V, 4A
38. In an a.c. circuit V and I are given by
 $V = 100 \sin (100 t)$ volts
 $I = 100 \sin (100t + \pi/3)$ mA
 The power dissipated in the circuit is
 (1) 10^4 watt (2) 10 watt
 (3) 2.5 watt (4) 5.0 watt
39. In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is-
 (1) $Q/2$ (2) $Q/\sqrt{3}$ (3) $Q/\sqrt{2}$ (4) Q
40. A radiation of energy ' E ' falls normally on a perfectly reflecting surface. The momentum transferred to the surface is (C = Velocity of light):-
 (1) $\frac{2E}{C}$ (2) $\frac{2E}{C^2}$ (3) $\frac{E}{C^2}$ (4) $\frac{E}{C}$
41. The speed of electromagnetic radiation in vacuum is:-
 (1) $\mu_0 \epsilon_0$ (2) $\sqrt{\mu_0 \epsilon_0}$ (3) $\frac{1}{\mu_0 \epsilon_0}$ (4) $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$
42. A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of water in the capillary tube is 5g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is :-
 (1) 2.5 g (2) 5.0 g (3) 10 g (4) 20 g

43. When water flows at a rate Q through a tube of radius r placed horizontally, a pressure difference p develops across the ends of the tube. If the radius of the tube is doubled and the rate of flow halved, the pressure difference will be :-
 (1) $8p$ (2) p (3) $\frac{p}{8}$ (4) $\frac{p}{32}$

44. A homogeneous solid cylinder of length L ($L < H/2$), cross-sectional area $A/5$ is immersed such that it floats with its axis vertical at the liquid-liquid interface with length $L/4$ in the denser liquid as shown in the figure. The lower density liquid is open to atmosphere having pressure P_0 . Then, density D of solid is given by:-

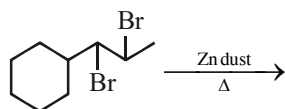


- (1) $\frac{5}{4}d$ (2) $\frac{4}{5}d$ (3) $4d$ (4) $\frac{d}{5}$
45. In determination of young's modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is '1m'. The curve between extension and stress is depicted then Young's modulus of wire will be :-



- (1) $2 \times 10^9 \text{ N/m}^2$ (2) $1 \times 10^9 \text{ N/m}^2$
 (3) $2 \times 10^{10} \text{ N/m}^2$ (4) $1 \times 10^{10} \text{ N/m}^2$

46. Product of following reaction is :-



- (1) (2)
 (3) (4)

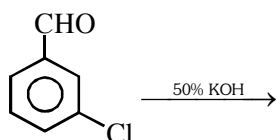
47. Phenol and benzoic acid are distinguished by :-

- (1) Lucas reagent (2) Victor Meyer test
 (3) Caustic soda (4) Sodium bicarbonate

48. Decarboxylation occurs with maximum rate in:

- (1) CH_3COOH (2) $\text{C}_6\text{H}_5\text{-COOH}$
 (3) $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$ (4) $\text{CH}_3\text{COCH}_2\text{COOH}$

49. Predict the products in the given reaction.



- (1) +
 (2) +
 (3) +
 (4) +

50. Of the following, which is the product formed when cyclohexanone undergoes aldol condensation followed by heating ?

- (1) (2)
 (3) (4)

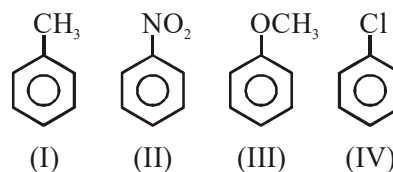
51. Which of the following reactions is appropriate for converting acetamide to methanamine ?

- (1) Hoffmann hypobromamide reaction
 (2) Stephens reaction
 (3) Gabriels phthalimide synthesis
 (4) Carbylamine reaction

52. Which of the following does not reduce Tollen's reagent :-

- (1) CH_3CHO (2) HCOOH
 (3) $\text{C}_6\text{H}_5\text{NHOH}$ (4) $\text{C}_6\text{H}_5\text{NH}_2$

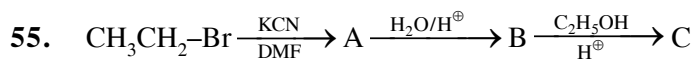
53. Correct reactivity order for electrophilic substitution reaction :



- (1) I > II > III > IV (2) III > I > IV > II
 (3) III > IV > I > II (4) I > III > IV > II

54. Which of the following compound gives foul smell with chloroform in presence of a base :

- (1) Ph-NH_2 (2) Ph-NH-CH_3
 (3) $(\text{CH}_3)_2\text{NH}$ (4) $\text{CH}_3\text{-N(CH}_3)_2$



Product C is :

- (1) An ester (2) An alcohol
 (3) An aldehyde (4) Carboxylic acid

56. Which of the following is not an addition polymer?

- (1) Teflon (2) PVC
(3) Dacron (4) BUNA-S

57. Which of the following is biodegradable polymer?

- (1) Nylon-2-Nylon-6 (2) Polyethene
(3) Nylon-6,6 (4) Bakelite

58. Find incorrect statement :

- (1) Each protein has different primary structure
(2) Secondary structure of protein consists of α -Helix and β -pleated sheet
(3) In denaturation of protein primary structure is destroyed
(4) In denaturation of protein secondary and tertiary structure is destroyed

59. Which of the following is non-reducing sugar ?

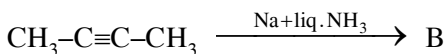
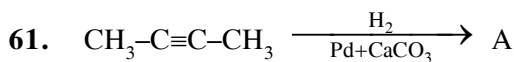
- (1) Maltose (2) Sucrose (3) Lactose (4) Fructose

60. Compare rate of S_N1 reaction :

- (I) $\text{Ph}-\text{CH}_2-\text{Br}$ (II) $\text{Ph}-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$

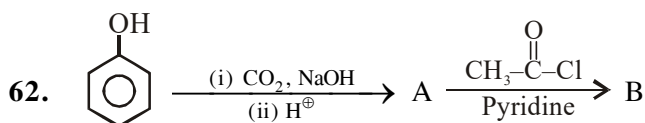
- (III) $\text{Ph}-\underset{\text{Ph}}{\text{CH}}-\text{Br}$ (IV) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$

- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{III} > \text{II} > \text{I} > \text{IV}$
(3) $\text{III} > \text{I} > \text{II} > \text{IV}$ (4) $\text{I} > \text{III} > \text{II} > \text{IV}$



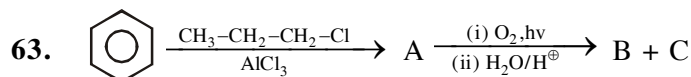
Products A and B are :

- (1) Geometrical isomers
(2) Identical
(3) Position isomers
(4) Chain isomers

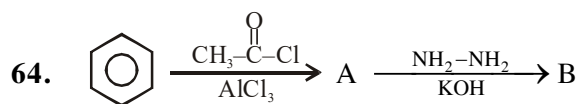
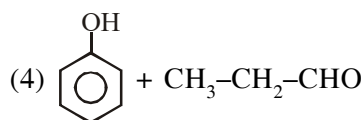
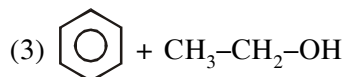
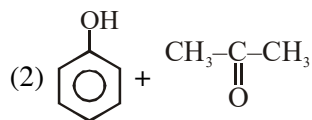
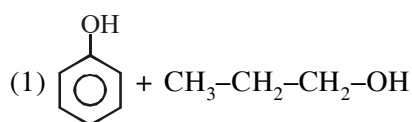


Product B is :

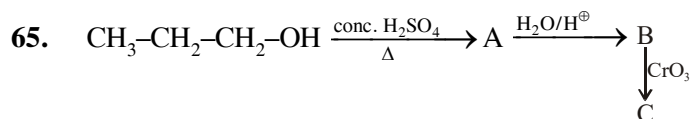
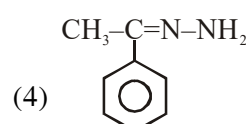
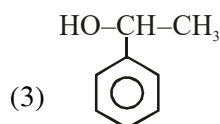
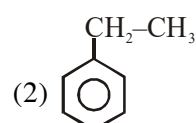
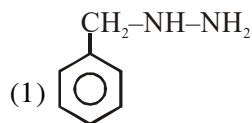
- (1) Acetyl salicylic acid
(2) Oil of wintergreen
(3) Phenyl salicylate
(4) Acetophenone



Products B and C are :



Product B is :



Product 'C' is :

- (1) $\text{CH}_3-\text{CH}_2-\text{CHO}$
(2) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$
(3) $\text{CH}_3-\text{CH}_2-\text{COOH}$
(4) $\text{CH}_3-\text{CH}_2-\text{COOH}$

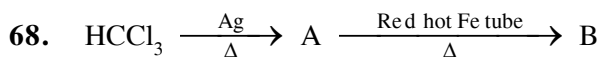
66. Which of the following show tautomerism ?

- (1) H-CHO (2) Ph-CHO



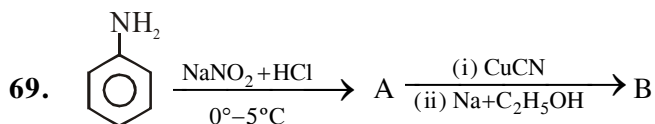
67. Gabriel phthalimide synthesis is good method for preparation of :

- (1) $\text{Ph}-\ddot{\text{N}}\text{H}_2$ (2) $\text{CH}_3-\text{NH}-\text{CH}_3$
(3) $\text{Ph}-\text{CH}_2-\text{NH}_2$ (4) $\text{CH}_3-\text{N}(\text{CH}_3)-\text{CH}_3$



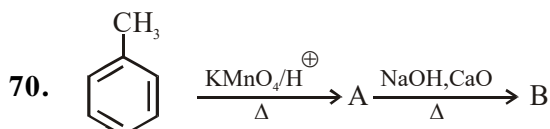
B is :

- (1)
(2) $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}_2$
(3) $\text{H}_2\text{C}=\text{CH}_2$
(4) $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}-\text{C}\equiv\text{CH}$



Product B is :

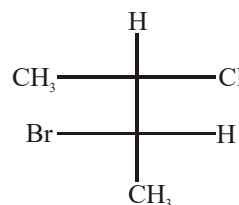
- (1) Ph-CN (2) Ph-CH₂OH
(3) Ph-CH₂NH₂ (4) Ph-NH₂



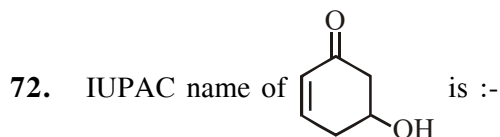
Product B is :-

- (1)
(2)
(3)
(4)

71. Absolute configuration of the given compound is :-



- (1) 2R, 3R (2) 2R, 3S
(3) 2S, 3R (4) 2S, 3S



- (1) 5-Hydroxycyclohex-2-en-1-one
(2) 3-Hydroxycyclohex-4-en-1-one
(3) 5-ketocyclohex-3-en-1-ol
(4) 3-ketocyclohex-4-en-1-ol

73. Chloroamphenicol is an :-

- (1) Antibiotic narrow spectrum
(2) Antibiotic limited spectrum
(3) Antibiotic broad spectrum
(4) Antiseptic

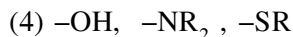
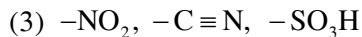
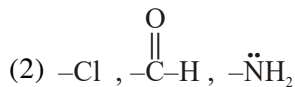
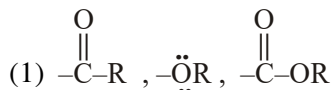
74. Which of the following is Narcotic analgesic.

- (1) Paracetamol
(2) Aspirin
(3) Morphine
(4) Chloroquine

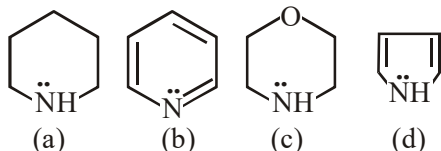
75. Which Alkyl group having maximum +I effect.

- (1) $-\text{CH}_3$
(2) $-\text{CH}_2-\text{CH}_3$
(3) $-\text{CH}(\text{CH}_3)_2$
(4) $-\text{C}(\text{CH}_3)_3$

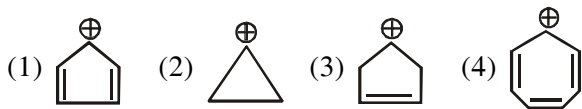
76. Which of following group show only -M effect (mesomeric effect)



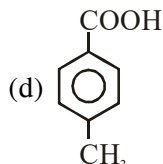
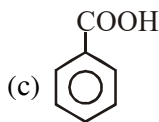
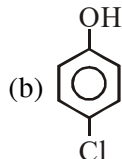
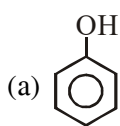
77. Basic strength order is :-



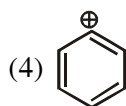
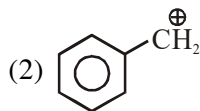
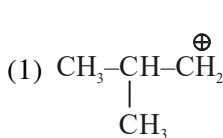
78. The most stable carbocation is



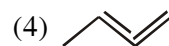
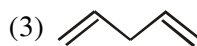
79. Acidic strength order is :



80. Which of the following will show hyperconjugation:



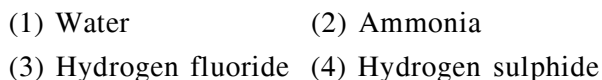
81. Which compound having more H.O.H. (Heat of Hydrogenation)



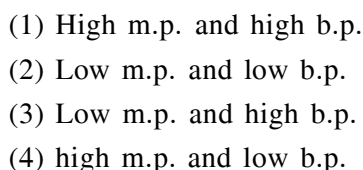
82. Maximum no. of hydrogen bonds formed by a water molecule in ice is



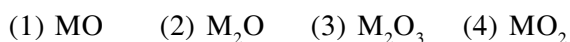
83. Strongest hydrogen bond is shown by :



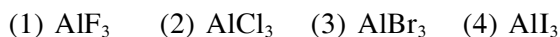
84. As compared to covalent compounds electrovalent compounds generally possess



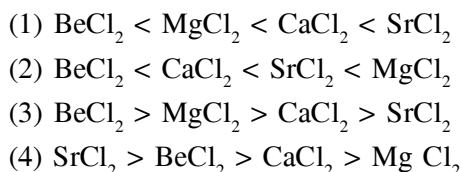
85. The electronic configuration of metal M is $1s^2 2s^2 2p^6 3s^1$. The formula of its oxide will be:



86. The most covalent halide is:-



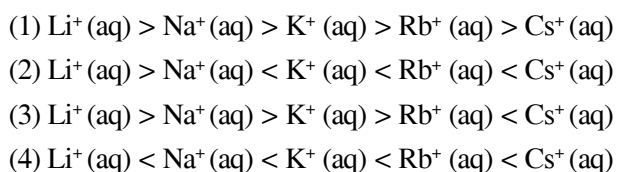
87. Correct order of covalent character of alkaline earth metal chloride in



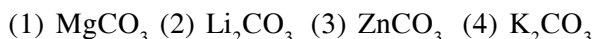
88. Which of the compound is least soluble in water



89. Ionic conductances of hydrated M^+ ions are in the order -



90. Which of the following does not give an oxide on heating -



- 91.** Cnidoblast is not found in :
 (1) *Hydra* (2) *Meandrina*
 (3) *Obelia* (4) *Fasciola*
- 92.** Metameric segmentation is not found in :
 (1) *Periplaneta* (2) *Nereis*
 (3) *Amphioxus* (4) *Holothuria*
- 93.** Mammary gland is not found in :
 (1) *Echidna* (2) *Balaenoptera*
 (3) *Macaca* (4) *Neophron*
- 94.** Find the incorrect match :
- | | Column-I
Animal | Column-II
Excretory structure |
|-----|----------------------------|--|
| (1) | <i>Ascaris</i> | Renette cell |
| (2) | Scorpion | Coxal gland |
| (3) | Siliver fish | Dermal branchiae |
| (4) | <i>Hyla</i> | Mesonephric kidney |
- 95.** Find the incorrect match :
- | | Column-I
Phylum | Column-II
Structures |
|-----|----------------------------|---------------------------------|
| (1) | Porifera | Choanocytes |
| (2) | Platyhelminthes | Flame cells |
| (3) | Mollusca | Mantle |
| (4) | Echinodermata | Water canal system |
- 96.** Most abundant cartilage in an adult human is :
 (1) Elastic cartilage
 (2) Hyaline cartilage
 (3) White fibrous cartilage
 (4) Calcified cartilage
- 97.** Simple cuboidal epithelium is not found in :
 (1) Thyroid follicles
 (2) PCT of nephron
 (3) Retina of eye
 (4) Myometrium of uterus
- 98.** All of the following are example of ductless gland except :
 (1) Pineal gland (2) Thyroid gland
 (3) Adrenal gland (4) Sudoriferous gland

- 99.** Haversian system is not found in bone of :
 (1) *Macropus* (2) *Rattus*
 (3) *Psittacula* (4) *Elephas*
- 100.** All of the following are example of dense connective tissue except :
 (1) Tendon (2) Ligament
 (3) Perichondrium (4) Ear pinna
- 101.** Find the incorrect statement related to gastric glands :
 (1) Secretion of HCl
 (2) Secretion of gastric juice which is acidic in nature
 (3) Contains peptic cell which mainly secretes castle intrinsic factor
 (4) Secretes prorennin and pepsinogen
- 102.** Most abundant agranulocyte in human blood is:
 (1) Neutrophils (2) Monocytes
 (3) Lymphocytes (4) Eosinophils
- 103.** Clotting protein like prothrombin and fibrinogen are synthesized by :
 (1) Thymus (2) Liver
 (3) Spleen (4) Bone marrow
- 104.** Agglutinin-A present on :
 (1) Platelets (2) Neutrophils
 (3) Erythrocytes (4) Basophils
- 105.** In human being heart beat rate not directly depends on :
 (1) Sex (2) Age
 (3) Size of person (4) BMR
- 106.** Semilunar valves open during :
 (1) Atrial systole (2) Ventricular diastole
 (3) Ventricular systole (4) Joint diastole
- 107.** In all of the following condition heart beat rate increases except :
 (1) Secretion of adrenal modullary hormone
 (2) Hypersecretion of thyroxine
 (3) Increase activity of parasympathetic nervous system
 (4) Weak heart
- 108.** Total number of teeth present in upper jaw of a 6 year old child is :
 (1) 12 (2) 16 (3) 14 (4) 10

109. Find the correct match :

	Column-I	Column-II
(1)	Enamel	Secreted by ameloblast
(2)	Mucosal epithelium in jejunum	Simple brush border columnar epithelium
(3)	Brunner's gland	Submucosa of duodenum
(4)	Crypts of Lieberkuhn	Muscularis of Ileum

110. Which of the following hydrolase unable to hydrolysed glycosidic bond ?

- (1) Lactase
- (2) Sucrase
- (3) Elastase
- (4) Maltase

111. Trypsinogen can be converted into trypsin by the action of :

- (1) Enterokinase only
- (2) Trypsin only
- (3) Pepsin
- (4) Both 1 and 2

112. Partial pressure of oxygen in pulmonary artery is :

- (1) 95 mm Hg
- (2) 104 mm Hg
- (3) 40 mm Hg
- (4) 46 mm Hg

113. Surfactant is secreted by :

- (1) Type-I pneumocytes
- (2) Type-II pneumocytes
- (3) Dust cells
- (4) Satellite cell

114. Which of the following lung volume or lung capacity can directly measured by spirometer ?

- (1) FRC
- (2) TLC
- (3) RV
- (4) VC

115. Total amount of CO₂ released by 700 ml deoxygenated blood into lungs is :

- (1) 140 ml
- (2) 105 ml
- (3) 28 ml
- (4) 40 ml

116. Find the incorrect match :

	Column-I	Column-II
(1)	Jaundice	Increase level of bile salt in blood
(2)	Emphysema	Surface area of lungs decreases
(3)	Mumps	Infectious parotitis
(4)	Myasthenia gravis	Autoimmune disorder of neuromuscular junction

117. Ultrafiltration is (Related to excretory system of human)

- (1) Active and non-selective process
- (2) Active and selective process
- (3) Passive and selective process
- (4) Passive and non-selective process

118. Find the incorrect statement related to juxtaglomerular cells :

- (1) JG cells found in arteriole of kidney
- (2) JG cells secretes erythropoietin
- (3) JG cells secretes renin
- (4) JG cells activated during hypertension

119. Find the incorrect match :

	Column-I	Column-II
(1)	Land snail	Uricotelic
(2)	Birds	Ureotelic
(3)	Tadpole larva of frog	Ammonotelic
(4)	Spider	Book lungs

120. After completion of counter current mechanism the osmolarity of human urine at site of papillary duct is :

- (1) 300 mOsm/L
- (2) 600 mOsm/L
- (3) 1200 mOsm/L
- (4) 900 mOsm/L

- 121.** The suture present between parietal and temporal bone is :
- (1) Coronal suture
 - (2) Squamous suture
 - (3) Lambdoidal suture
 - (4) Sagittal suture
- 122.** Head of femur and acetabulum of pelvic girdle form a joint, it is a type of :
- (A) Ball and socket joint
 - (B) Synovial joint
 - (C) Fibrous joint
 - (D) Diarthroses
- Answer code :-
- (1) B and C
 - (2) C and D
 - (3) A and C
 - (4) A, B and D
- 123.** The bone unrelated to axial skeleton is :
- (1) Zygomatic
 - (2) Hyoid
 - (3) Sternum
 - (4) Patella
- 124.** Vertebrosteral ribs are connected to sternum with the help of :
- (1) Fibrous cartilage
 - (2) Hyaline cartilage
 - (3) Calcified cartilage
 - (4) Elastic cartilage
- 125.** Find the incorrect one related to a cardiac muscle fibres :
- (1) Uninucleated
 - (2) Intercalated disc
 - (3) Striated
 - (4) Voluntary

- 126.** Epimysium and perimysium are absent in :
- (A) Skeletal muscles
 - (B) Smooth muscles
 - (C) Cardiac muscles

Answer code :-

- (1) A
- (2) A and B
- (3) B and C
- (4) A, B, C

- 127.** All of the following disorder are related to skeletal system except :

- (1) Osteoporosis
- (2) Osteoarthritis
- (3) Muscular dystrophy
- (4) Rheumatoid arthritis

- 128.** Find the incorrect match :

	Column-I	Column-II	Column-III	Column-IV
(1)	Earthworm	Nephridia	Cutaneous respiration	Tube with in tube body plan
(2)	<i>Pila</i>	Organ of Bojanus	Branchial respiration	Protostomes
(3)	<i>Antedon</i>	Excretory system absent	Water canal system	Deuterostomes
(4)	<i>Ascidia</i>	Hemichordate	Retrogressive metamorphosis	Deuterostomes

- 129.** Human heart and kidney originates from :

- (1) Embryonic ectoderm
- (2) Embryonic mesoderm
- (3) Embryonic endoderm
- (4) Extra embryonic mesoderm

- 130.** Find the correct match :

	Column-I	Column-II	Column-III	Column-IV
(1)	PCT	Maximum reabsorption	Brush border columnar epithelium	Found only in cortical nephron
(2)	Loop of Henle	Related to concentration of urine	Simple squamous and simple cuboidal epithelium	Found in all type of kidney i.e. mesonephric Metanephric
(3)	Cervical vertebrae	7 in mammals	1st cervical is Atlas	7th is known as Axis
(4)	Pancreatic juice	Complete digestive juice	Secreted by exocrine part of Pancreas	Released into Duodenum

131. Find the incorrect match :

	Column-I	Column-II	Column-III	Column-IV
(1)	Zygomatic bone	1 pair in human	Part of axial skeleton	Found in face
(2)	Femur	1 pair in human	Part of appendicular skeleton	Found in legs
(3)	Biceps	1 pair in human	Voluntary muscles	Found in arms
(4)	Ureter	1 pair in human	Lined by stratified squamous keratinized epithelium	Found in abdominal cavity

132. Find the correct match :

	Column-I	Column-II	Column-III
	Cells	Function	System
(1)	Eosinophils	Kill bacteria by secreting perform	Circulatory system
(2)	G-cell	Secretes Gastrin	Gastrointestinal system
(3)	Pancreatic acinar cell	Secretes pancreatic juice	Neuroendocrine system
(4)	Macula densa cell	Secretes Erythropoietin	Excretory system

133. The use of drugs like _____ and _____ quickly reduce the symptoms of allergy :-

- (1) Histamin, Adrenalin, Steroids
- (2) Histamine nor Adrenalin, Steroids
- (3) Anti Histamine, Adrenalin, Steroids
- (4) Anti Histamine, antioibotics, Steroids

134. The foetus also receives some antibodies from their mother through the placenta during pregnancy, it is example of :-

- (1) Artificial passive immunity
- (2) Natural active immunity
- (3) Artificial active immunity
- (4) Natural passive immunity

135. Out of these how many statements are correct

- (a) Entamoeba histolytica causes amoebiasis
- (b) The round worm cause a slowly developing acute inflammation of the organs
- (c) Microsporium, Trichophyton and epidermophyton are not responsible for ring worms
- (d) Haemophilus influenzae are responsible for the pneumonia

- (1) a, b
- (2) a,c,d
- (3) a,d
- (4) All are correct

136. Sub sequent encounter with the same pathogen elicits a highly intensified _____ response :-

- (1) Secondary immune response
- (2) Primary immune response
- (3) Anamnestic response
- (4) 1 & 3 both

137. Which one of the following pairs of disease can spread through blood transfusion :-

- (1) AIDS & Malaria
- (2) AIDS & Syphilis
- (3) Diabetes Mellitus & Syphilis
- (4) Hepatitis-B & AIDS

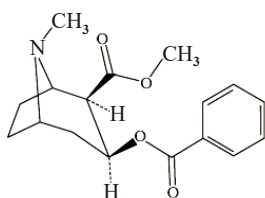
138. Find out the correct match of following hormone with their source and functions ?

	Hormone	Source	Function
(1)	Adrenaline	Ectodermal adrenal part	Increase glucose level in blood & stimulate protein synthesis
(2)	Cortisol	Adrenal Zona fasciculata layer	Stimulates lipolysis and gluconeogenesis
(3)	Thyroxine	Endodermal	Hyperthyroidism during pregnancy causes defective development of growing baby called cretinism
(4)	Aldosterone	Zona Reticularis	Maintenance of osmotic pressure

139. Which of the following hormones is related with generation of secondary messenger ?

- (1) Adrenalin and thyroxine
- (2) Estrogen and Iodothyronin
- (3) Thyroxine and insuline
- (4) Adrenaline and FSH

140. The A,B,C and D figures below are related to drugs of abuse potential .Select the option giving correct description of these figures:-



(A)



(B)



(C)



(D)

- (1) Figure A--Cannabis sativa used to form marijuana , charas and hashish
- (2) Figure B- Cannabis sativa giving cocaine which inhibits transport of dopamine
- (3) Figure C-Papaver somniferum producing compounds acting through opioid receptors
- (4) Figure D- Atropa belladonna having sedative properties

141. Which of the following is a vector borne viral disease?

- (1) Chikungunya
- (2) Common cold
- (3) Filariasis
- (4) Malaria

142. Which of the following statement is false :-

- (1) Heroin is chemically diacetylmorphine
- (2) Excessive dosage of cocaine causes hallucination.
- (3) Receptors of opioids are present only in Brain.
- (4) Cannabinoids have effects on cardiovascular system of the body.

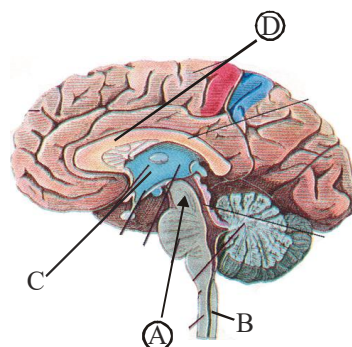
143. Read the following statements (A-D)

- (A)Dependence leads the patient to ignore all social norms in order to get sufficient funds to satiate his/her needs.
- (B)Adolescence is accompanied by several biological and behavioral changes.
- (C)Curiosity need for adventure and excitement and experimentation constitute common causes, which motivate youngsters towards drug and alcohol abuse.
- (D)Unstable or unsupportive family structure and peer pressure are associated with drug / alcohol abuse.

How many of the above statement are true.

- (1) 3
- (2) 4
- (3) 2
- (4) 1

144. Find out the correct labellings for following diagram :-

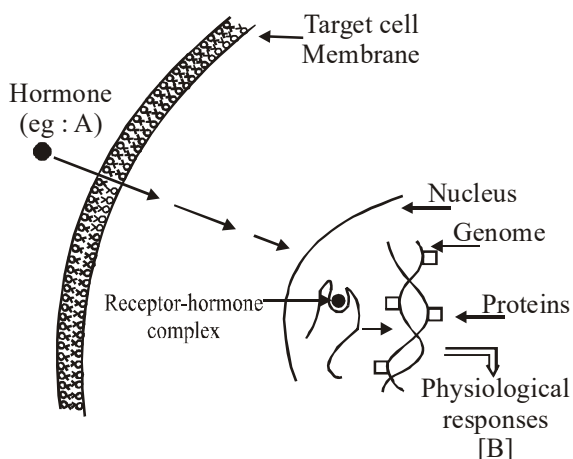


- (1) (A)Osmotic centre (B)Oculomotor nerve
(C)Motor signalling (D)Tract of nerve fibres
- (2) (A)Pneumotaxic centre (B)Optic lobes
(C)Olfactory signaling (D) Involuntary reflex
- (3) (A)Limb muscle (B) Involuntary reflex control
(C) Sensory signaling (D) Tract of nerve fibre
- (4) (A) Visual reflex (B)Involuntary control of voluntary muscles
(C) Olfactory signaling(D)Largest commissure

145. A patient is suffering from high fever, stomach pain, Headache, Loss of appetite, Constipation intestinal perforation. Which amongst following can be used as a diagnostic test:

- (1) WIDAL Test
- (2) ELISA Test
- (3) Mantoux test
- (4) Shick test

146. Which main property of acquired immunity is the basis of concept of *vaccination* ?
 (1) Specificity
 (2) Diversity
 (3) Discrimination b/w self & non-self
 (4) Memory
147. Which part of brain gets affected by alcohol and as a result of that the person cannot maintain his balance and walking is disturbed?
 (1) Cerebrum (2) Cerebellum
 (3) Medulla Oblongata (4) Pons
148. Semicircular canals in internal ear are concerned with:
 (1) Hearing
 (2) Balance of body
 (3) Transmission of sound vibrations to cochlea
 (4) Balance of pressure on either side of the eardrum
149. Identify the diagrammatic representation of the mechanism of hormone action given below.



Select the option with incorrect matching of hormone [A] with physiological response [B] regarding the above mechanism.

	A	B
(1)	Estrogen	Uterine tissue growth
(2)	Progesterone	Supports pregnancy and stimulate physical growth of mammary gland
(3)	Cortisol	Inflammatory response
(4)	Thyroxin	Calorigenic effect

150. Which of the following is the correct arrangement of cells from inside to outside in our retina ?
 (1) Photoreceptor cells, bipolar cells and ganglion cell
 (2) Photoreceptor cells, ganglion and biopolar cells
 (3) Ganglion cells, bipolar and photoreceptor cells
 (4) Ganglion cells, photoreceptor and bipolar cells
151. Which of the following group of hormone is not correctly matched with their origin and functions ?

	Hormone	Origin	Function
(1)	Glucocorticoids	Adrenal cortex	Stimulate gluconeogenesis and lipolysis
(2)	PTH	Parathyroid gland	Bone demineralisation
(3)	Oxytocin	Anterior pituitary	Vigorous contraction of uterus during parturition
(4)	Insulin	β cell of pancreas	Stimulates glycogenesis

152. Match column-I with column-II and choose the correct option :

Column-I		Column-II	
A	Adrenal	I	Graves disease
B	Pancrease	II	Diabetes Insipidus
C	Pituitary	III	Acromegali
D	Thyroid	IV	Addison disease
		V	Diabetes mellitus

- (1) A-(III); B-(V); C-(IV); D-(I)
 (2) A-(IV); B-(II); C-(III); D-(I)
 (3) A-(IV); B-(V); C-(III); D-(I)
 (4) A-(I); B-(II); C-(IV); D-(III)

153. Which of the following is/are not correct ?

(i)	Scotopic vision	Function of Rods
(ii)	Daylight vision	Function of Rods
(iii)	Colour vision	Function of Cones
(iv)	Twilight vision	Function of Cones

- (1) i and ii (2) ii and iv
(3) i and iv (4) ii and iii

154. First origin of life occurred in :

- (1) Water
(2) Air
(3) At 800°C temp.
(4) In vapour state

155. Marsupial mole, koala, bandicoot & wombat are example of :

- (1) Placental mammals
(2) Convergent evolution
(3) Adaptive radiation
(4) 1 & 3 both

156. Which reptile went back into water to evolve into fish like reptiles ?

- (1) *Ichthyosaurs*
(2) *Calotes*
(3) *Hemidactylus*
(4) Alligator

157. The biggest land reptile was :

- (1) *Pelycosaurus*
(2) *Calotes*
(3) *Tyrannosaurus*
(4) *Bangarus*

158. Select the true statements :

- (1) *Ramapithecus* and *Dryopithecus* were existing about 50 million years ago
(2) *Ramapithecus* was more man like while *Dryopithecus* was more ape like
(3) *Ramapithecus* was more ape like while *Dryopithecus* was more man-like
(4) 1 & 2 both

159. The story of evolution of modern man in respect to brain & language appears as :

- (1) Convergent evolution
(2) Divergent evolution
(3) Parallel evolution
(4) All

160. Highest brain capacity present in :

- (1) *Australopithecus*
(2) *Homo habilis*
(3) *Homo erectus*
(4) Neanderthal man

161. When more than one adaptive radiations appear in an isolated geographical area to form similar organism then it is :-

- (1) Selection
(2) Convergent evolution
(3) Use and disuse
(4) Atavism

162. Who gave the theory of genetic equilibrium ?

- (1) Lamarck
(2) Hugo deVries
(3) Hardy-Weinberg
(4) Darwin

163. First mammals were :

- (1) Human (2) Apes
(3) Monkeys (4) Shrews

164. Proper burial of dead bodies for the first time started by which pre historic man ?

- (1) Java man (2) *Homo habilis*
(3) Neanderthal man (4) *Australopithecines*

165. Who lived in near east and central Asia between 1,00,000 - 40,000 years back?

- (1) *Homo erectus* (2) *Homo habilis*
(3) Neanderthal man (4) *Australopithecines*

- 166.** When reptiles came down which animals took over the earth ?
 (1) Birds
 (2) Dinosaurs
 (3) Mammals
 (4) Amphibians
- 167.** Which vertebrates evolved into the first amphibians?
 (1) Jawless fishes
 (2) Lobefins
 (3) Salamanders
 (4) *Ichthyosaurus*
- 168.** The naturalist who came about with the same ideas that of the Darwin :
 (1) Alfred Nobel
 (2) Alfred Wallace
 (3) Malthus
 (4) H. Khorana
- 169.** According to Darwin, the term fitness refers to :
 (1) Social fitness
 (2) Mental fitness
 (3) Physical fitness
 (4) Reproductive fitness
- 170.** Origin of species was written by :-
 (1) Malthus
 (2) Khorana
 (3) Darwin
 (4) Haldane
- 171.** Connecting link between reptiles and mammals is :-
 (1) Prototheria
 (2) Peripatus
 (3) Coelocanth
 (4) Neopilina
- 172.** In *Funeria*. Meiosis takes place in :
 (1) Spore mother cell
 (2) Zygote
 (3) Prothallus
 (4) Archegonium
- 173.** In the list given below. How many members exhibit haplontic life cycle?
Ulothrix, Fucus, Polysiphonia, Spirogyra, Rhizopus, Mucor, Pteridium, Sphagnum
 (1) Two
 (2) Five
 (3) Four
 (4) Three
- 174.** Select the incorrect statement :
 (1) Viroids are naked RNA particles with high molecular weight
 (2) The most notable disease caused by Prions is mad cow disease
 (3) W.M stanley showed that viruses could be crystallised
 (4) Mycorrhiza is symbiotic association in between fungal component and roots of higher plants.
- 175.** Free living gametophyte is found in :
 (1) *Selaginella*
 (2) *Mangifera*
 (3) *Gnetum*
 (4) *Adiantum*
- 176.** Select the mismatched pair :
 (1) Prothallus- *Salvinia*
 (2) Protonema-*Funeria*
 (3) Coralloid root- *Cycas*
 (4) Mycorrhiza-*Pinus*
- 177.** Viruses are regarded as "Contagium vivum Fluidum" by :
 (1) Ivanowsky
 (2) Beijerinck
 (3) Nawaschin
 (4) Antone von leeuvenhoek

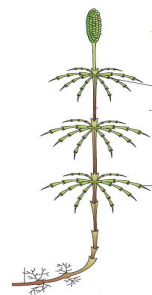
178. Identify the given diagram and select the correct option :



(A)



(B)



(C)



(D)

	A	B	C	D
1	<i>Sphagnum</i>	<i>Salvinia</i>	<i>Equisetum</i>	<i>Ginkgo</i>
2	<i>Selaginella</i>	<i>Ginkgo</i>	<i>Funeria</i>	<i>Salvinia</i>
3	<i>Funeria</i>	<i>Salvinia</i>	<i>Equisetum</i>	<i>Ginkgo</i>
4	<i>Funeria</i>	<i>Selaginella</i>	<i>Equisetum</i>	<i>Pinus</i>

179. If number of chromosomes in Leaves of *Pinus* is 30. The number of chromosomes in its endosperm will be :

- (1) 30
- (2) 20
- (3) 15
- (4) 60

180. Match the column I with column-II and choose the correct answer :

	Column-I		Column-II
A	Lycopsida	i	<i>Adiantum</i>
B	Sphenopsida	ii	<i>Lycopodium</i>
C	Psilopsida	iii	<i>Equisetum</i>
D	Pteropsida	iv	<i>Psilotum</i>

A	B	C	D
(1) iv	i	ii	iii
(2) ii	iv	iii	i
(3) ii	iii	iv	i
(4) iv	iii	ii	i

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SPACE FOR ROUGH WORK

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REVISED

SPARK COURSE TEST PLAN (2019-20)

DATE	DAY	TEST No.	SYLLABUS GUIDELINES	PATTERN
19-01-2020	Sunday	PART TEST 1	Test 1 (100%)	NEET
02-02-2020	Sunday	PART TEST 2	Test 1 (20%) + Test 2 (80%)	NEET
16-02-2020	Sunday	PART TEST 3	Test 1 (10%) + Test 2 (20%) + Test 3 (70%)	NEET
01-03-2020	Sunday	PART TEST 4	Test 2 (10%) + Test 3 (20%) + Test 4 (70%)	NEET
08-03-2020	Sunday	SEMI MAJOR	Test 1, 2, 3, 4	NEET
15-03-2020	Sunday	PART TEST 5	Test 3 (10%) + Test 4 (20%) + Test 5 (70%)	NEET
29-03-2020	Sunday	PART TEST 6	Test 4 (10%) + Test 5 (20%) + Test 6 (70%)	NEET
16-04-2020	Thursday	PART TEST 7	TEST 5 (10%)+TEST 6 (20%) + TEST 7 (70%)	NEET
19-04-2020	Sunday	MAJOR TEST-1	Full Syllabus (AIOT)	NEET
22-04-2020	Wednesday	MAJOR TEST-2	Full Syllabus (Clubbed with ALLEN KOTA Students)	NEET

SPARK COURSE - I (SEMI MAJOR)

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Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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PATTERN / TYPE : NEET / SEMI MAJOR
ANSWER KEY
PRE-MEDICAL 2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	1	3	4	2	3	4	1	3	1	1	3	1	4	3	1	3	3	1	4	1	3	3	3	4	2	1	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	3	2	1	2	3	3	3	1	4	3	4	1	1	2	4	4	1	1	1	4	2	1	1	3	1	3	2	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	1	2	2	2	3	3	1	3	1	3	1	3	3	4	3	4	4	4	1	4	1	3	1	2	4	3	4	4	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	4	3	4	2	4	4	3	4	3	3	2	3	3	3	3	4	4	3	4	3	2	4	3	1	4	4	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	2	4	3	3	3	2	4	4	2	3	4	3	4	4	2	4	3	1	3	2	3	1	4	2	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	2	1	3	1	3	2	3	4	2	3	4	3	3	3	2	2	4	3	1	1	3	1	4	1	2	3	3	3

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SPARK COURSE - I

TARGET : PRE-MEDICAL 2020

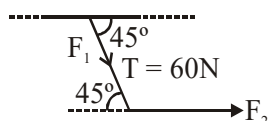
Test Type : **SEMI MAJOR**

Test Pattern : **NEET(UG)**

TEST DATE : 08 - 03 - 2020

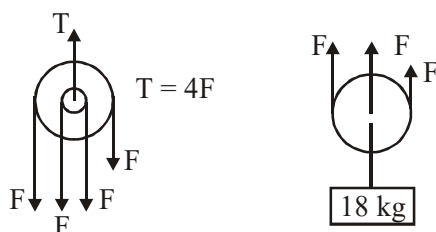
HINT - SHEET

1.



$$F_1 = F_2 = T \cos 45^\circ = 60 \times \frac{1}{\sqrt{2}} = 30\sqrt{2} \text{ N}$$

2.



$$3F = 18g$$

$$F = 6g = 60 \text{ N}$$

$$T = 4F = 240 \text{ N}$$

3.

$$Mg = 50 \Rightarrow M = 5 \text{ kg}$$

$$W_{ap} = m(g - a)$$

4.

Area of power time graph = work

Area is +ve thus work increases

5.

$$V_A = \sqrt{5g\ell}$$

$$V_B = \sqrt{3g\ell}$$

$$V_C = \sqrt{g\ell}$$

6.

At height h from ground P.E = mgh,

$$v = \sqrt{2g(H - h)}$$

$$KE = 2PE$$

$$\frac{1}{2}mv^2 = 2mgh$$

$$\frac{1}{2}m(2g(H - h)) = 2mgh$$

$$\text{Solving, } h = H/3, v = 2\sqrt{gH/3}$$

7.

$$\vec{L}_i = \vec{L}_f$$

$$mvL = \left(\frac{ML^2}{3} + mL^2 \right) \omega$$

9.

$$Fv = P \Rightarrow m \left(\frac{vdv}{dx} \right) v = P$$

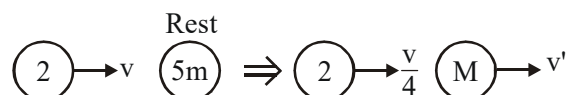
$$\int_0^v v^2 dv = \int_0^x \frac{P}{m} dx$$

10.

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh = \frac{1}{2}m(R^2\omega^2) + \frac{1}{2} \left(\frac{MR^2}{2} \right) \omega^2$$

11.



$$e = \frac{v' - \frac{v}{4}}{v - 0} = 1$$

$$\Rightarrow v' = \frac{5v}{4}$$

Apply conservation of linear momentum

$$2v = \frac{2v}{4} + m \times \left(\frac{5v}{4} \right)$$

$$\Rightarrow m = 1.2 \text{ kg}$$

$$12. |\vec{A}| = \sqrt{2^2 + 4^2 + 5^2} = \sqrt{45}$$

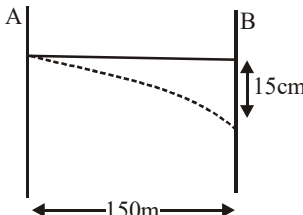
$$\cos \alpha = \frac{A_x}{A} = \frac{2}{\sqrt{45}}$$

$$\cos \beta = \frac{A_y}{A} = \frac{4}{\sqrt{45}}$$

$$\cos \gamma = \frac{A_z}{A} = \frac{-5}{\sqrt{45}}$$

13. Slope of x-t graph gives velocity

14. Area under a-t graph gives change in velocity.

15. 

$$R = u \sqrt{\frac{2h}{g}}$$

$$150 = u \sqrt{\frac{2 \times 15}{10 \times 100}}$$

$$u = 500\sqrt{3} \text{ m/s}$$

$$16. \vec{v}_c = -5\hat{j}$$

$$\vec{v}_{b/c} = -2\sqrt{6}\hat{i}$$

$$\vec{v}_b = \vec{v}_{b/c} + \vec{v}_c$$

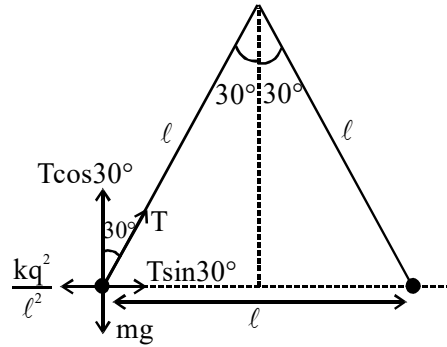
$$\vec{v}_b = -2\sqrt{6}\hat{i} - 5\hat{j}$$

$$\text{so magnitude} = \sqrt{(2\sqrt{6})^2 + (5)^2} = \sqrt{49} = 7 \text{ m/s}$$

$$17. \vec{E}_1 \perp \vec{E}_2 \text{ and } E_1 = E_2 = \frac{\sigma}{2\epsilon_0} = \frac{\sqrt{2}}{2} \times 10^{-3}$$

$$\text{So } E = \frac{2 \times 10^{-3}}{2} = 10^{-3} \text{ N/C}$$

18.



$$T \cos 30^\circ = mg \text{ \& } T \sin 30^\circ = \frac{kq^2}{\ell^2}$$

$$\Rightarrow \tan 30^\circ = \frac{kq^2}{\ell^2 mg}$$

$$\Rightarrow m = \frac{kq^2 \sqrt{3}}{\ell^2 g} = 0.15 \text{ kg}$$

19. Electric potential decreases in the direction of electric field.

$$20. E_x = -\frac{dV}{dx} = -\frac{d(4x^2)}{dx} = -8x$$

$$(E_x)_{1,0,2} = -8 \times 1 = -8 \text{ V/m}$$

$$\vec{E} = E_x \hat{i}$$

$$\vec{E} = -8\hat{i} \text{ V/m}$$

21. Zero (No change in potential energy)

22. Capacitor behaves as short circuit. So

$$\text{equivalent resistance} = \frac{3R}{2}. \text{ Hence } I = \frac{2V}{3R}$$

23. For isothermal process, work done per mole, i.e.,

$$PV = RT$$

$$\Rightarrow P = \frac{RT}{V}$$

$$\therefore W = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{RT}{V} dV = RT \log_e \frac{V_2}{V_1}$$

24. Coefficient of performance

$$K = \frac{T_2}{T_1 - T_2}$$

$$\Rightarrow 5 = \frac{(273 - 13)}{T_1 - (273 - 13)} = \frac{260}{T_1 - 260}$$

$$\Rightarrow 5T_1 - 1300 = 260 \Rightarrow 5T_1 = 1560$$

$$\Rightarrow T_1 = 312 \text{ K} \rightarrow 39^\circ \text{C}$$

25. The two-third of heat produced goes into the bullet.

$$\text{So, } m \times s \times \Delta\theta = \frac{2}{3} \times \frac{1}{2} mu^2$$

$$\Delta\theta = \frac{u^2}{3 \times s} = \frac{180 \times 180}{3 \times 120} = 90^\circ\text{C}$$

26. $(\Delta Q)_P = nC_P \Delta T \Rightarrow 207 = 1 \times C_P \times 10$

$$\Rightarrow C_P = 20.7 \frac{\text{joule}}{\text{mol-K}}. \text{ Also } C_P - C_V = R$$

$$\Rightarrow C_V = C_P - R = 20.7 - 8.3 = 12.4 \frac{\text{joule}}{\text{mol-K}}$$

$$\text{So, } (\Delta Q)_V = nC_V \Delta T = 1 \times 12.4 \times 10 = 124\text{J}$$

27. According to Newton's law of cooling, rate of cooling is given by

$$\left(\frac{-dT}{dt} \right) = \frac{4\sigma A \epsilon T_0^3}{ms} (T - T_0)$$

where, c is specific heat of material.

$$\left(\frac{-dT}{dt} \right) \propto \frac{1}{s}$$

i.e., rate of cooling varies inversely as specific heat. From the graph, for A rate of cooling is larger. Therefore, specific heat of A is smaller.

28. $1.2 = x\ell$

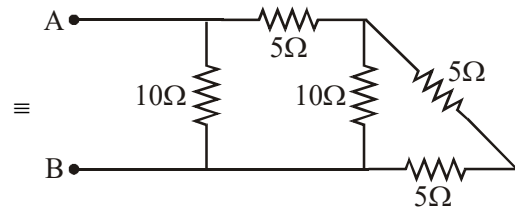
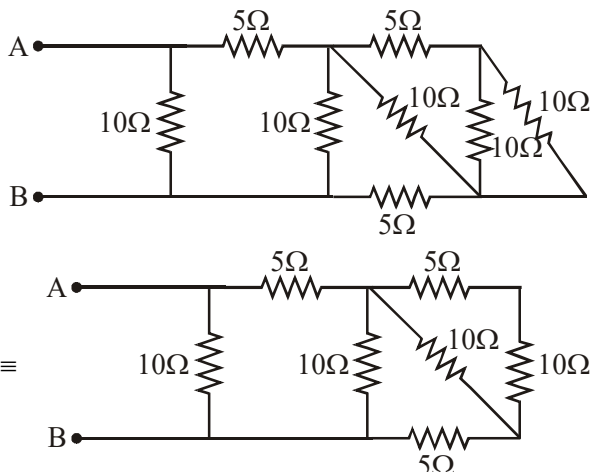
$$x = \frac{V_{AB}}{L} = \frac{i \times R}{L} = \frac{1/5 \times 10}{100 \times 10^{-2}} = \frac{10}{5} = 2$$

$$i = \frac{3}{5+10} = \frac{3}{15} = \frac{1}{5}\text{A}$$

$$\text{so } 1.2 = x\ell \Rightarrow 1.2 = 2 \times \ell$$

$$\ell = \frac{6}{10} \times 100 = 60\text{ cm}$$

- 29.



30. When S_1 is closed only $I_1 = \frac{E}{R+6R} = \frac{E}{7R}$

$$\text{When } S_2 \text{ is closed only } I_2 = \frac{E}{R+3R} = \frac{E}{4R}$$

When S_1 and S_2 both closed

$$I_3 = \frac{E}{R+2R} = \frac{E}{3R}$$

32. $q = N \frac{\Delta\phi}{R}$

$$N = 1$$

$$\Delta\phi = qR$$

$$q = \text{Area under } i-t \text{ curve}$$

$$\Delta\phi = \frac{1}{2} \times 4 \times 0.1 \times 10$$

$$\Delta\phi = 2$$

34. $P_{\text{out}} = 0.9 P_{\text{in}}$

$$P_{\text{out}} = 2700$$

$$i_{\text{out}} v_{\text{out}} = 2700$$

$$v_{\text{out}} = \frac{2700}{6} = 450\text{ V}$$

$$P_{\text{in}} = i_{\text{in}} v_{\text{in}}$$

37. Resonance cond.

$$V = V_R$$

42. $hr = h' \times 2r$

$$h' = \frac{h}{2}$$

$$v_1 = \pi R^2 h$$

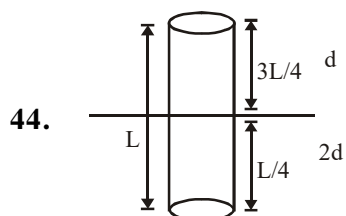
$$v_2 = 2v_1$$

$$m_2 = 2m_1$$

$$v_2 = 2\pi R^2 h$$

43.
$$\frac{Q}{\text{rate of flow}} = \frac{\pi \Delta P r^4}{8 \eta L}$$

$$\Delta P = \frac{Q \times 8 \eta L}{\pi r^4}$$



$$Mg = F_{\text{buoyant}}$$

$$DL \frac{A}{5} = 2d \frac{L}{4} \frac{A}{5} + d \times \frac{3L}{4} \times \frac{A}{5}$$

$$D = \frac{d}{2} + \frac{3d}{4}$$

$$D = \frac{5d}{4}$$

45.
$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Stress} \times L}{\Delta L}$$

$$\frac{1}{Y} = \frac{\Delta L}{\text{Stress}} \times \frac{1}{L}$$

Slope of ΔL and stress graph

138. NCERT Pg # 333 & 335

139. NCERT Pg # 338

140. NCERT XIIth Pg # 158-159 Fig. 8.7, 8.10, 8.8, 8.11

142. NCERT 12th Page # 158, 159 (E)

143. NCERT 12th Page # 160, 161 (E)

145. NCERT XII (E) pg. # 146 last para
(H) pg.# 158, 1st para

146. NCERT (E) pg.# 152 3rd para
(H) pg.# 164, second para

149. NCERT (XI) Pg. # 339

150. NCERT (XI) Pg. # 323

151. NCERT XI Eng. pg.#335, 334, 332, 336
Hin. pg.#335, 334, 332, 336

152. NCERT page NO. 346

153. NCERT (XI) Pg. # 324